

Notes: Liu & Palmer (JFE, 2026)

Implicit Extrapolation and the Beliefs Channel of Investment Demand

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1 Big picture

- **Question.** Do stated expectations fully capture the beliefs that investors use when making investment decisions?
- **Core answer.** No. Investors extrapolate from past local housing returns not only when reporting return forecasts, but also again when choosing how much to invest.
- **Main concept.** The paper distinguishes **explicit extrapolation**, where past returns affect stated expected returns, from **implicit extrapolation**, where past returns directly affect investment choices even after controlling for stated expectations.
- **Empirical setting.** The authors use the New York Fed Survey of Consumer Expectations housing module from 2015–2021, where respondents report perceived past home-price appreciation, expected future home-price appreciation, risk aversion, subjective confidence, and investment choices.
- **Main investment outcome.** Respondents choose how to allocate a hypothetical \$1,000 between a local housing-market fund and a savings account paying 2%.
- **Main result.** Perceived past home-price appreciation predicts the housing-fund share even after controlling for forecasted future appreciation, risk tolerance, confidence, and other controls.
- **Magnitude.** Allowing for implicit extrapolation makes the effect of past returns on investment roughly twice as large as what would be inferred from the path “past returns → stated forecasts → investment.”
- **Mechanism.** Many investors are less confident in their forecasts than in their recollection of past returns. These investors rely more on past returns at the investment stage.
- **Survey evidence.** Respondents who rely on expected returns cite fundamentals and expectations. Respondents who rely on past returns cite consistent trends, real data, and uncertainty.
- **Economic takeaway.** Survey-elicited beliefs can be incomplete sufficient statistics for investment decisions. Investors may report one forecast but still use other belief inputs when choosing portfolios.

2 Data and measurement

2.1 Survey source and sample

- The paper uses the housing module of the New York Fed Survey of Consumer Expectations.
- The main sample covers survey years 2015–2021.
- The survey asks respondents about local home-price growth, expectations over the next year, confidence in their answers, and real-estate investment decisions.
- Respondents are asked demographic and financial-literacy questions used as controls and for heterogeneity analysis.

Interpretation: The survey is valuable because it observes three objects jointly for the same person: perceived past returns, stated expected future returns, and investment choices.

2.2 Investment outcome

The main outcome is the share of a hypothetical \$1,000 allocated to a local housing-market fund:

$$Y_{i,t} = \text{share invested in local housing-market fund by respondent } i \text{ in year } t. \quad (1)$$

The alternative is a savings account paying 2% per year.

Interpretation: This question creates a clean risky-asset share outcome. It avoids relying only on real housing transactions, which are infrequent and constrained by borrowing, moving, and search frictions.

2.3 Belief variables

The paper measures:

- **Perceived past HPA:** respondent’s recalled home-price appreciation in the local area over the previous 12 months.
- **Forecasted HPA:** respondent’s expected local home-price appreciation over the next 12 months.
- **Actual past HPA:** objective local home-price appreciation over the previous year.
- **Perception gap:** perceived past HPA minus actual past HPA.

Interpretation: The paper focuses on subjective experience. Perceived past returns matter because investors may act on their memories and interpretations, not only on objective price indices.

2.4 Confidence and risk measures

- Respondents report confidence in their perceived past home-price estimate.
- Respondents also report confidence in their expected future home-price forecast.
- The confidence gap is the difference between confidence in forecasts and confidence in past returns.
- Risk tolerance is measured on a survey scale and is used to control for risk preferences.
- Some specifications flexibly control for the respondent’s subjective distribution of future returns, including probabilities of positive and negative home-price changes.

Interpretation: Confidence is central to the mechanism: when investors trust remembered past returns more than forecasts, they may weight past returns more heavily in decisions.

2.5 Additional outcomes

The paper also studies:

- probability of buying a nonprimary residence;
- probability of buying a primary residence;
- probability of buying an investment property;
- self-reported reasons for relying more on expected returns or past returns.

Interpretation: These outcomes test whether implicit extrapolation is specific to the hypothetical portfolio question or also appears in broader housing-related behavior and stated decision rationales.

3 Models and empirical specifications

3.1 Benchmark portfolio-choice model

The paper uses the classic risky-asset share formula to frame the empirical test:

$$\phi_t = \frac{E_t[r_{t+1}] - r_f}{\alpha \sigma_t^2}. \quad (2)$$

- $E_t[r_{t+1}]$ is expected return.
- r_f is the risk-free return.
- α is risk aversion.
- σ_t^2 is perceived return variance.

Interpretation: In this benchmark, past returns affect investment only through expected returns, risk, or risk aversion. Once those are controlled for, past returns should have no direct effect.

3.2 Main investment regression

The main empirical model can be written as:

$$Y_{i,t} = \beta_0 + \beta_1 \hat{r}_{i,t} + \beta_2 \hat{E}_i[r_{i,t+1}] + X'_{i,t} \gamma + \varepsilon_{i,t}, \quad (3)$$

where $\hat{r}_{i,t}$ is perceived past home-price appreciation and $\hat{E}_i[r_{i,t+1}]$ is stated expected future home-price appreciation.

Interpretation: Explicit extrapolation is captured by the effect of perceived past returns on stated forecasts. Implicit extrapolation is captured by $\beta_1 > 0$ in an investment regression that already controls for stated expected returns.

3.3 Belief-formation regression

The paper first estimates how past returns affect stated expected returns:

$$\hat{E}_i[r_{i,t+1}] = \alpha + \delta \hat{r}_{i,t} + X'_{i,t} \Gamma + u_{i,t}. \quad (4)$$

Interpretation: A positive δ documents explicit extrapolation. It shows that investors incorporate perceived past home-price growth into future return forecasts.

3.4 Confidence mechanism regression

To test whether confidence explains implicit extrapolation, the paper interacts forecasted and past returns with the confidence gap:

$$Y_{i,t} = \alpha + \beta_1 \hat{E}_i[r_{i,t+1}] + \beta_2 \hat{r}_{i,t} + \beta_3 \hat{E}_i[r_{i,t+1}] \times ConfGap_i + \beta_4 \hat{r}_{i,t} \times ConfGap_i + \beta_5 ConfGap_i + X'_{i,t} \gamma + \varepsilon_i. \quad (5)$$

Interpretation: If respondents are more confident in forecasts than in past returns, forecasted returns should matter more and past returns should matter less. This predicts $\beta_3 > 0$ and $\beta_4 < 0$.

3.5 Forward-looking versus backward-looking respondents

Respondents are classified based on whether they say they rely more on stated expected returns or past returns when making investment decisions. The paper estimates separate and interacted regressions:

$$Y_i = \alpha + \beta_1 Forecast_i + \beta_2 Past_i + \beta_3 Forward_i + \beta_4 Forecast_i \times Forward_i + \beta_5 Past_i \times Forward_i + X'_i \gamma + \varepsilon_i. \quad (6)$$

Interpretation: Forward-looking respondents should load more on forecasts and less on past returns. Backward-looking respondents should load more on past returns.

3.6 Factor reweighting test

The paper compares the weight of belief factors in forecast regressions and investment regressions. Forecasted returns are explained by perceived past returns, expected rent growth, and expected inflation. Investment is then explained by the same factors.

Interpretation: If forecasts were sufficient statistics for all belief inputs, the investment regression should not reweight belief factors. The paper finds that past returns receive more weight in investment than in forecasts, while other factors receive less weight.

4 Identification and empirical design

4.1 What identifies implicit extrapolation

- The key comparison is between the effect of perceived past returns on stated forecasts and the effect of perceived past returns on investment conditional on stated forecasts.
- The survey observes both the belief domain and the decision domain for the same respondent.
- The main specifications control for stated expected home-price growth, risk tolerance, confidence, demographics, and probabilities of future price changes.

Interpretation: The identifying idea is not that past returns are randomly assigned. It is that, under standard belief-channel logic, stated forecasts should summarize the relevant belief component for investment. Direct predictive power of past returns after conditioning on forecasts reveals missing belief information or decision-stage weighting.

4.2 Addressing risk and wealth channels

- Past home-price growth could affect investment by changing perceived risk, not expected return.
- It could also affect homeowners' housing wealth and therefore risk tolerance or leverage.
- The paper uses risk tolerance, subjective probabilities of future price changes, and flexible controls for the distribution of expected returns.
- Results remain when adding these richer controls.

Interpretation: The direct effect of past returns is not easily explained by measured risk aversion or subjective downside-risk beliefs.

4.3 Addressing survey-measurement concerns

- Survey forecasts may be noisy, which could mechanically leave residual predictive power for past returns.
- The paper discusses and tests survey-noise, collinearity, and framing explanations.
- Robustness exercises show that implicit extrapolation survives using alternative samples and additional controls.

Interpretation: The paper's mechanism evidence goes beyond a measurement-error story by showing that confidence gaps and self-reported rationales predict how investors reweight past returns versus forecasts.

4.4 What the paper cannot fully prove

- The investment experiment is hypothetical, although the paper also examines real housing-related intentions.
- Perceived past returns are subjective and may reflect other unobserved beliefs.
- Confidence is self-reported and may itself be endogenous.
- The setting is housing investment, so extrapolation in other asset classes may differ.

Interpretation: The paper’s strength is that it observes stated forecasts and investment decisions jointly, allowing a direct test of whether forecasts are sufficient statistics for decisions.

5 Tables and figures

Visuals are directly cropped from the paper and grouped compactly to improve reading speed.

5.1 Figure 1 and Table 1: survey design and summary statistics

Read:

- Figure 1 shows the investment allocation question and the subjective confidence question.
- Respondents allocate \$1,000 between a local housing fund and a savings account paying 2%.
- Respondents also state confidence intervals for perceived past home-price changes.
- Table 1 shows that the sample averages 51 years old, is 53% male, and is 76% homeowner.
- Mean forecasted HPA is 3.81 percentage points, while mean perceived past HPA is 4.73 percentage points.
- The average housing-fund allocation is 57.26 percentage points.

Economic message: The survey is designed to observe beliefs, confidence, and investment allocations in the same respondent, which is essential for testing implicit extrapolation.

I. Investment question

Consider a situation where you have to decide how to invest \$1,000 for one year. You can choose between two possible investments.

The first is a fund that invests in your local housing market, and pays an annual return equal to the growth in home prices in your area. The second is a savings account that pays 2% interest per year.

What proportion of the \$1,000 would you invest in:

(Please note: The numbers need to add up to 100.)

The housing market fund	 %
The savings account	 %
TOTAL: 0	

II. Subjective uncertainty question

You stated over the past 12 months, the value of a typical home in your zip code increased by 9%. How certain are you in your answer?

- ☐ (Completely certain) I am certain that the value of a typical home in my zip code increased by 9% over the past 12 months.
- ☐ (Very certain) I am certain that the value of a typical home in my zip code changed between an increase of 7.0% and an increase of 11.0% over the past 12 months.
- ☒ (Somewhat certain) I am certain that the value of a typical home in my zip code changed between an increase of 3.0% and an increase of 15.0% over the past 12 months.
- ☐ (Little certain) I am certain that the value of a typical home in my zip code changed between a decrease of 1.0% and an increase of 19.0% over the past 12 months.
- ☐ (Very uncertain) I am not at all certain that the value of a typical home in my zip code changed between a decrease of 1.0% and an increase of 19.0% over the past 12 months.

Fig. 1. Investment and subjective confidence survey questions.

(a) Fig. 1: survey questions.

Table 1
Summary statistics: 2015–2020 sample.

	Response Count	Mean	Std. Dev.
<i>I. Individual characteristics</i>			
Age (years)	7,065	51.22	19.04
Male Indicator	7,064	0.53	0.50
Minority Indicator	7,056	0.16	0.37
Married Indicator	7,066	0.65	0.48
Homeowner Indicator	7,025	0.76	0.43
College Graduate Indicator	7,064	0.57	0.50
1(Household Income $\geq$ \$100K)	6,998	0.29	0.45
1(Liquid Savings $\geq$ \$75K)	6,630	0.39	0.49
Numeracy Score (0-5)	7,065	4.05	1.05
Risk Tolerance (1-10)	7,066	4.45	2.24
<i>II. Beliefs and investment actions</i>			
Forecasted HPA in the Next 12 months (p.p.)	7,056	3.81	4.85
Perceived HPA in the Past 12 months (p.p.)	7,053	4.73	6.18
Confidence in Perceived Past Returns (1-5)	7,053	3.20	0.93
Confidence in Forecasted Return (1-5)	984	2.96	1.03
Actual HPA in the Past 12 months (p.p.)	7,020	5.35	4.03
Perception Gap (p.p.)	7,007	4.91	4.44
Share Invested in a Housing Fund (p.p.)	3,015	57.26	34.26
Probability of Moving within 3 years	7,050	0.30	0.34
Probability of Buying a Primary Residence	4,999	0.67	0.33
Probability of Buying an Investment Property	7,049	0.10	0.18

Notes: Table reports means, standard deviations, and counts of individual responses used in the empirical analysis. Numeracy is coded between 1 and 5, based on the number of correct answers to 5 questions testing numerical literacy. Risk tolerance is coded from 1 (risk averse) to 10 (risk loving). Confidence level of past home-price growth estimate is coded from 1 (not all confident) to 5 (very confident). Perception Gap is the absolute value of the difference between a respondent's perception of last year's home-price growth in their zip code and zip-code-level returns estimated from CoreLogic's repeat-sales index. Share invested in a housing fund is asked in both 2015 and 2020 and represents the share of a hypothetical \$1000 investment allocated by the respondent to an index of local housing market returns instead of a savings account with a 2% annual yield. Likelihood of buying a primary residence is asked to respondents who report an over-5% probability of moving within 3 years.

(b) Table 1: summary statistics.

5.2 Tables 2–3 and Figure 2: explicit and implicit extrapolation

Read:

- Table 2 shows explicit extrapolation: perceived past returns strongly predict forecasted returns.
- A one percentage point higher perceived past return predicts about 24–29 basis points higher forecasted return across specifications.
- Table 3 shows implicit extrapolation: perceived past returns predict investment allocations even when forecasted returns are controlled for.
- In Table 3, forecasted returns and perceived past returns both have positive and significant coefficients in the same investment regression.
- Figure 2 visualizes the same point: the housing-fund share rises with perceived past returns both unconditionally and conditional on expected returns.

Economic message: Stated forecasts do not fully summarize the belief inputs used in investment decisions. Investors continue to load on remembered past returns after their stated expectations are controlled for.

Table 2

The effect of perceived past returns on belief formation.

Dependent Variable:	Forecasted Returns			
	(1)	(2)	(3)	(4)
Perceived Past Returns	0.29*** (0.013)	0.28*** (0.014)	0.25*** (0.013)	0.24*** (0.014)
Forecasted Rent Growth			0.15*** (0.011)	0.15*** (0.011)
Forecasted Inflation			0.07*** (0.02)	0.07*** (0.02)
Individual Controls		✓		✓
Fundamentals			✓	✓
Observations	6,993	6,993	6,993	6,993
R-squared	0.139	0.163	0.202	0.222

Notes: Dependent variable is surveyed expected house price appreciation over the next year. Perceived past returns are respondent's estimate of home-price appreciation in their zip code over the past year. One percentage point is denoted as 1. Individual controls include binary indicators for owning a home, numeracy, ethnicity, gender, marital status, education, labor force status, census region, age, and logs of household income, equity in home, liquid savings, personal debt, a dummy for consulting websites about home prices in the past 12 months, and a dummy for receiving questions in a percentage-change framing instead of a level framing, a dummy that equals 1 if respondent reports being confident in their recall of past home price changes (i.e., answers 4 or more on a 1–5 scale, where 5 is very confident), a dummy that equals 1 if respondent

(a) Table 2: belief formation.

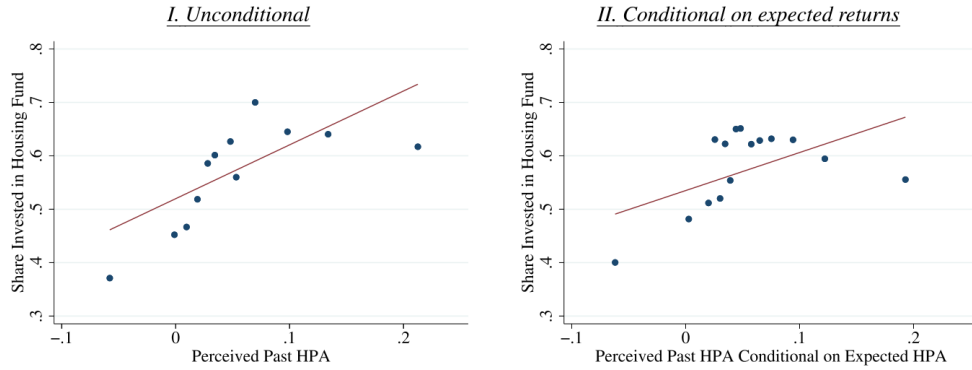
Table 3

Effects of forecasted and past returns on investment.

Dependent Variable:	Housing Fund Share (on a 0–100 scale)					
	(1)	(2)	(3)	(4)	(5)	(6)
Forecasted Returns	1.30*** (0.13)		0.88*** (0.15)	1.23*** (0.13)		0.93*** (0.14)
Perceived Past Returns			1.01*** (0.10)	0.71*** (0.11)	0.85*** (0.10)	0.54*** (0.11)
Confidence in Past Returns				5.48*** (1.29)	5.44*** (1.30)	5.35*** (1.29)
Above-median Risk Aversion				−9.63*** (1.29)	−9.45*** (1.30)	−9.38*** (1.29)
Male				6.44*** (1.27)	6.68*** (1.26)	6.52*** (1.26)
Homeowner				1.11 (1.54)	0.11 (1.55)	0.67 (1.54)
Individual Controls				✓	✓	✓
Observations	2,963	2,963	2,963	2,963	2,963	2,963
R-squared	0.033	0.035	0.047	0.129	0.123	0.136

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). One percentage point is denoted as 1. Confident in past returns is a dummy that equals 1 if respondent reports being confident in their recall of past home price changes (i.e., answers 4 or more on a 1–5 scale, where 5 is very confident). Above-median risk aversion is a dummy that equals 1 if respondent reports a 4 or less (on 1–10 scale) to a question about willingness to take risks in financial matters, where 10 is very willing. Individual controls are controlled in columns 4 to 6. For definitions of these controls, see notes to Table 2. Robust standard errors in parentheses. Significant at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(b) Table 3: investment regressions.



Risky asset shares and perceived past returns.

Figure presents binned scatter plots for the share of an \$1000 investment in the housing fund versus the perceived past home-price growth unconditional I and after first partialling out individual-level forecasted home-price growth in panel II. $N = 2966$.

5.3 Table 4 and Table 5: risk aversion and confidence

Read:

- Table 4 shows that risk tolerance is a strong predictor of housing-fund investment, but controlling for it does not remove the effect of perceived past returns.
- Forecasted returns and perceived past returns remain jointly important when probability-distribution controls and individual controls are included.
- Table 5 tests the confidence mechanism.
- The interaction between perceived past returns and confidence in forecasts relative to past returns is negative.
- As respondents become more confident in forecasts relative to past returns, they rely less on perceived past returns in investment decisions.

Economic message: Implicit extrapolation is not just omitted risk aversion. It is related to how confident investors are about forecasts versus remembered past returns.

THE FUND OF THIS INVESTMENT IN INVESTMENT DECISIONS.

Dependent Variable:	Housing Fund Share				
	(1)	(2)	(3)	(4)	(5)
Risk Tolerance (1-10)	3.70*** (0.28)	2.76*** (0.29)	2.74*** (0.29)		
Forecasted Returns		0.77*** (0.14)	0.54*** (0.15)	0.75*** (0.14)	0.54*** (0.15)
Perceived Past Returns			0.48*** (0.11)		0.46*** (0.11)
Confident in Past Returns		4.48*** (1.29)	4.40*** (1.29)	4.64*** (1.29)	4.56*** (1.29)
Probabilities Cubic		✓	✓	✓	✓
Individual Controls		✓	✓	✓	✓
Risk Tolerance Score FE × Year FE				✓	✓
Observations	2,963	2,963	2,963	2,963	2,963
R-squared	0.059	0.161	0.167	0.172	0.178

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). One percentage point is denoted as 1. Probabilities cubic is a vector of controls for a cubic polynomial for each respondent's stated probability that next year's returns fall in one of four ranges—see Section 2.1 for question

(a) Table 4: risk aversion.

Table 5
Effects of confidence on investment decision-making.

Dependent Variable:	Housing Fund Share (on a 0-100 scale)			
	(1)	(2)	(3)	(4)
Forecasted Returns	1.28*** (0.24)	0.87*** (0.34)	0.77*** (0.33)	0.20 (0.35)
Perceived Past Returns	0.53*** (0.17)	0.78*** (0.23)	0.82*** (0.24)	0.79*** (0.24)
Forecasted Returns × (Conf Forecast - Conf Past)	0.12 (0.24)	0.11 (0.23)	0.02 (0.21)	0.06 (0.21)
Perceived Past Returns × (Conf Forecast - Conf Past)	-0.56*** (0.17)	-0.52*** (0.17)	-0.46*** (0.17)	-0.47*** (0.16)
Confidence in Forecast Returns - Confidence in Past Returns	3.39* (1.86)	3.11* (1.88)	5.08*** (1.95)	4.23** (1.91)
Risk Tolerance (1-7)	6.58*** (0.72)	6.57*** (0.72)	5.11*** (0.78)	5.03*** (0.77)
Forecasted Returns × Treated		0.89* (0.46)	0.94** (0.45)	0.88** (0.44)
Perceived Past Returns × Treated		-0.55 (0.34)	-0.74** (0.34)	-0.71** (0.33)
Treated		1.62 (3.35)	2.05 (3.37)	3.01 (3.26)
Individual Controls			✓	✓
Distribution of Forecasted Returns				✓
Observations	925	925	925	925
R-squared	0.161	0.165	0.233	0.257

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). One percentage point is denoted as 1. Treated is a dummy for the treatment group, who receives the nudging question before reporting their investment choices. Conf Forecast - Past is the difference between reported confidence in the respondent's 1-year HPA forecast and reported confidence in 1-year perceived past HPA. For definitions of individual controls, see notes to Table 2. Controls for the distribution of forecasted returns is a vector of linear controls for a respondent's stated probability of future returns

(b) Table 5: confidence and decision-making.

5.4 Table 6 and Table 7: forward-looking versus backward-looking respondents Read:

- Table 6 splits respondents by whether they say they rely more on expected returns or past returns.
- Forward-looking respondents load strongly on forecasted returns and weakly on perceived past returns.
- Backward-looking respondents load strongly on perceived past returns and weakly on forecasted returns.
- In the full sample, the interaction of forecasted returns with the forward-looking indicator is positive, while the interaction of perceived past returns with the forward-looking indicator is negative.
- Table 7 shows factor reweighting: past returns matter both for forecasts and investment, while rent growth and inflation matter more for forecasts than for investment.

Economic message: Respondents' self-reported reliance on forecasts versus past returns matches their actual regression weights. This supports the interpretation that investors deliberately reweight belief factors at the decision stage.

Table 6
Investment decision factors by forward- and backward-looking.

Dependent Variable:	Housing Fund Share (on a 0-100 scale)		
Sample:	Forward-Looking	Backward-Looking	Full Sample
	(1)	(2)	(3)
Forecasted Returns	1.41*** (0.27)	0.42 (0.32)	0.50 (0.31)
Perceived Past Returns	0.19 (0.22)	1.16*** (0.25)	1.28*** (0.24)
Forward-Looking			7.52*** (2.70)
Forecasted Returns × Forward-Looking			0.91** (0.41)
Perceived Past Returns × Forward-Looking			-1.14*** (0.32)
Individual Controls	✓	✓	✓
Observations	772	613	1,385
R-squared	0.175	0.234	0.185

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). The sample in column 3 is all respondents in the 2020-2021 surveys that were asked whether they rely more on their perceived past (column 2 sample) or their stated expected returns (column 1 sample) when making investment decisions. Forward-looking is an indicator for respondents that reported relying more on their stated expected returns. One percentage point is denoted as 1. For definitions of individual

(a) Table 6: forward- and backward-looking respondents.

Table 7
Evidence for factor reweighting.

Dependent Variable:	Forecasted Returns	Housing Fund Share	
	(1)	(2)	(3)
Forecasted Returns			0.53*** (0.15)
Perceived Past Returns	0.29*** (0.02)	0.80*** (0.10)	0.49*** (0.11)
Forecasted Rent Growth	0.14*** (0.02)	0.07 (0.11)	-0.09 (0.11)
Forecasted Rate of Inflation	0.12*** (0.03)	-0.05 (0.15)	-0.17 (0.15)
Individual Controls	✓	✓	✓
Probabilities Cubic			✓
Observations	2,963	2,963	2,963
R-squared	0.276	0.144	0.170

Notes: Dependent variable in column 1 is surveyed expected house price appreciation over the next year. Dependent variable in columns 2-3 is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). For definitions of individual controls, see notes to Table 2. Probabilities cubic is a vector of controls for a cubic polynomial for each respondent's stated probability that next year's returns fall in one of four ranges—see Section 2.1 for question framing. Robust standard errors in

(b) Table 7: factor reweighting.

5.5 Tables 8–9: heterogeneity

Read:

- Table 8 shows heterogeneity by renter/homeowner status, college education, and income.
- Perceived past returns remain important across many subgroups, including homeowners, non-college respondents, and income groups.
- Table 9 shows heterogeneity by age, gender, numeracy, and whether respondents checked housing websites.
- Forecasted returns are stronger for some financially sophisticated groups, while perceived past returns remain important in many groups.

Economic message: Implicit extrapolation is broad but not uniform. Financial sophistication and information acquisition change how respondents weight forecasts and past returns.

Table 8

Heterogeneity in investment decision-making, economic characteristics.

Dependent Variable: Sample	Housing Fund Share (on a 0-100 scale)					
	Renter (1)	Owner (2)	Non-College (3)	College (4)	Income ≤\$75K (5)	Income >\$75K (6)
Forecasted Returns	0.46* (0.24)	0.64*** (0.19)	0.32 (0.20)	0.85*** (0.23)	0.42** (0.18)	0.91*** (0.29)
Perceived Past Returns	0.17 (0.21)	0.63*** (0.14)	0.59*** (0.15)	0.45*** (0.17)	0.58*** (0.14)	0.40* (0.21)
Pr(HPA < 0%)	-0.15** (0.06)	-0.14*** (0.03)	-0.18*** (0.04)	-0.11*** (0.04)	-0.16*** (0.04)	-0.10** (0.05)
Pr(HPA < -5%)	0.02 (0.09)	0.01 (0.06)	0.06 (0.07)	-0.04 (0.07)	0.06 (0.06)	-0.09 (0.08)
Pr(HPA > 10%)	0.07 (0.07)	0.05 (0.04)	0.05 (0.05)	0.06 (0.06)	0.06 (0.05)	0.05 (0.06)
Individual Controls	✓	✓	✓	✓	✓	✓
Observations	734	2,229	1,263	1,700	1,633	1,305
R-squared	0.134	0.166	0.155	0.163	0.134	0.172

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). For definitions of individual controls, see notes to Table 2. Robust standard errors in parentheses. Significant at *p<0.10, **p<0.05, ***p<0.01.

Table 9

Heterogeneity in investment decision-making, other characteristics.

Dependent Variable: Sample	Housing Fund Share (on a 0-100 scale)							
	Age<50 (1)	Age≥50 (2)	Female (3)	Male (4)	Low Numeracy (5)	High Numeracy (6)	Did not Check (7)	Checked Website (8)
Forecasted Ret.	0.74*** (0.21)	0.43* (0.22)	0.43** (0.20)	0.79*** (0.22)	0.09 (0.23)	0.85*** (0.19)	-0.01 (0.27)	1.02*** (0.24)
Perceived Past Ret.	0.31* (0.16)	0.65*** (0.16)	0.52*** (0.15)	0.45*** (0.17)	0.54*** (0.18)	0.52*** (0.14)	0.67*** (0.20)	0.28 (0.18)
Pr(HPA < 0%)	-0.15*** (0.04)	-0.14*** (0.04)	-0.13*** (0.04)	-0.13*** (0.04)	-0.09* (0.06)	-0.15*** (0.03)	-0.08 (0.06)	-0.18*** (0.05)
Pr(HPA < -5%)	0.07 (0.07)	-0.02 (0.06)	0.06 (0.06)	-0.04 (0.07)	0.01 (0.09)	0.002 (0.06)	0.01 (0.09)	-0.01 (0.08)
Pr(HPA > 10%)	0.06 (0.06)	0.05 (0.05)	0.03 (0.05)	0.09* (0.05)	0.12* (0.06)	0.03 (0.05)	0.14* (0.08)	-0.01 (0.06)
Individual Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	1,377	1,586	1,398	1,565	779	2,184	744	1,287
R-squared	0.185	0.140	0.135	0.148	0.158	0.162	0.137	0.174

Notes: Dependent variable is the share of a \$1000 investment allocated by a respondent to an index of her local home-price appreciation (with the remainder allocated to a savings account earning 2%). For definitions of individual controls, see notes to Table 2. Robust standard errors in parentheses. Significant at *p<0.10, **p<0.05, ***p<0.01.

5.6 Figure 3: self-reported rationales

Read:

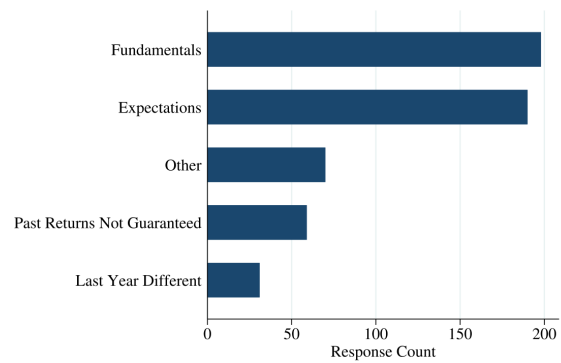
- Panel I shows why respondents rely more on expected returns. The dominant categories are fundamentals and expectations.
- Panel II shows why respondents rely more on past returns. The dominant categories are consistent trends and real data.
- Uncertainty is also an important reason for relying on past returns.

Economic message: The open-ended responses match the confidence mechanism. Many investors treat past returns as concrete, reliable data, while forecasts feel uncertain or abstract.

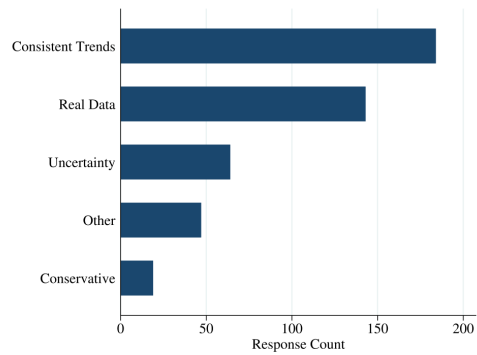
6 Short summary

- The paper studies whether stated expectations fully capture the beliefs driving investment decisions.
- The setting is housing investment expectations and decisions in the Survey of Consumer Expectations.
- Respondents report perceived past HPA, expected future HPA, confidence, and a housing-fund allocation.
- Perceived past returns predict stated expected returns, documenting explicit extrapolation.
- Perceived past returns also predict investment conditional on stated expected returns, documenting implicit extrapolation.
- The implicit component makes the total effect of past returns on investment much larger than what standard stated-belief regressions imply.

I. Reasons for relying on expected returns



II. Reasons for relying on past returns



Answers to investment decision factor rationale

- Risk aversion and subjective distribution controls do not eliminate the past-return effect.
- Confidence gaps help explain the mechanism: investors rely more on the signal they trust more.
- Respondents who call themselves forward-looking load more on forecasts; those who rely on past returns load more on past returns.
- Open-ended responses support the interpretation that some investors view past returns as concrete evidence and forecasts as uncertain.

7 Conclusion

7.1 Core conclusion

Liu and Palmer show that stated return forecasts are not always sufficient statistics for investment decisions. Investors often extrapolate from past returns twice: first when forming stated forecasts, and again when deciding how much to invest.

7.2 Economic intuition

A respondent may report a forecast based on several inputs, but when making a decision, may trust some inputs more than others. Past returns can feel tangible, memorable, and reliable. Forecasts can feel uncertain. When confidence in forecasts is low relative to confidence in past experience, investors overweight past returns in their actual investment choices.

7.3 Takeaway

The paper qualifies the beliefs channel of investment demand. Measuring stated expectations is valuable, but survey forecasts may not fully reveal the beliefs used in decisions. To understand investment behavior, researchers need to measure both forecasts and the confidence-weighted decision rules through which investors use them.